EE 671: VLSI DESIGN

PROJECT REPORT -2

Team Information

Group No.	Name	Roll No.
44	Amit Arvind Kale	25D0545
	Antriksh	25M1205
	Karunesh Kumar Tripathi	25D0529
	Tejas Sunita Anil Hande	25D0537

Problem Statement

Design and Implementation of a 4×4 (16 8-bit inputs) max pooling operation utilizing a 2×2 kernel.

Table of Contents

INTRODUCTION		4
MAX-POOLING DESIGN CONCEPT		6
PART– 1: PRE-SYNTHESIS		8
1.1	Output and gtkwave Waveforms	
a.	Output Transcript	
b.	Output Waveform	
c.	Output Waveform [Zoomed In]	
PART– 2: POST-SYNTHESIS		10
2.1	Area and Summary Report	
a.	Summary Report	
b.	Area Report	
c.	Gates Report	
d.	Power Report	
2.2	Output and gtkwave Waveforms	
a.	Output Transcript	
b.	Output Waveform	
c.	Output Waveform [Zoomed In]	
2.3	Summary Table	
PART– 3: PHYSICAL DESIGN		17
3.1	DRC, Antenna and Connection Report	
a.	DRC Check	
b.	Antenna Violation Check	
c.	Connectivity Report (Regular)	

d.	Connectivity Report (Special)	
3.2	Final Layout	
3.3	Post Route Timing, Area and Power Report	
a.	Area Report	
b.	Setup Time Report	
c.	Hold Time Report	
d.	Power Report	
3.4	Post Physical Design Output and gtkwave Waveforms	
a.	Output Transcript	
b.	Output Waveform	
c.	Output Waveform [Zoomed In]	
Table of Contributions		27

INTRODUCTION

Max Pooling

Max pooling is a simple but extremely useful operation in convolutional neural networks (CNNs). When a CNN processes an image, its early layers respond to important structures—edges, curves, textures, reflections, and other visual cues. These detections typically produce detailed feature maps that highlight where meaningful patterns appear in the image.

However, these feature maps can become large and dense. Not all of that information is equally important, and feeding every detail into deeper layers can make the network unnecessarily heavy and sensitive to noise. Max pooling helps address this by condensing each small region of a feature map into a single representative value.

The idea is straightforward: Take a small window—often 2×2—and select only the maximum value inside it. This value usually corresponds to the strongest activation, meaning the most dominant feature detected in that area. By repeating this across the entire feature map, max pooling keeps the important signals while discarding small variations or irrelevant noise.

This provides two major benefits:

1. **Simplification of data – The feature maps become smaller and easier for the network to process.**
2. **Stability – Small changes in the input image (like slight shifts, lighting differences, or noise) have less effect on the pooled output. The network still picks up the key visual cues.**

The figure below illustrates this idea. The left side shows how the model detects edges and patterns across the car's body and grille, while the right side shows how max pooling condenses these detections into a smoother, more compact feature map.

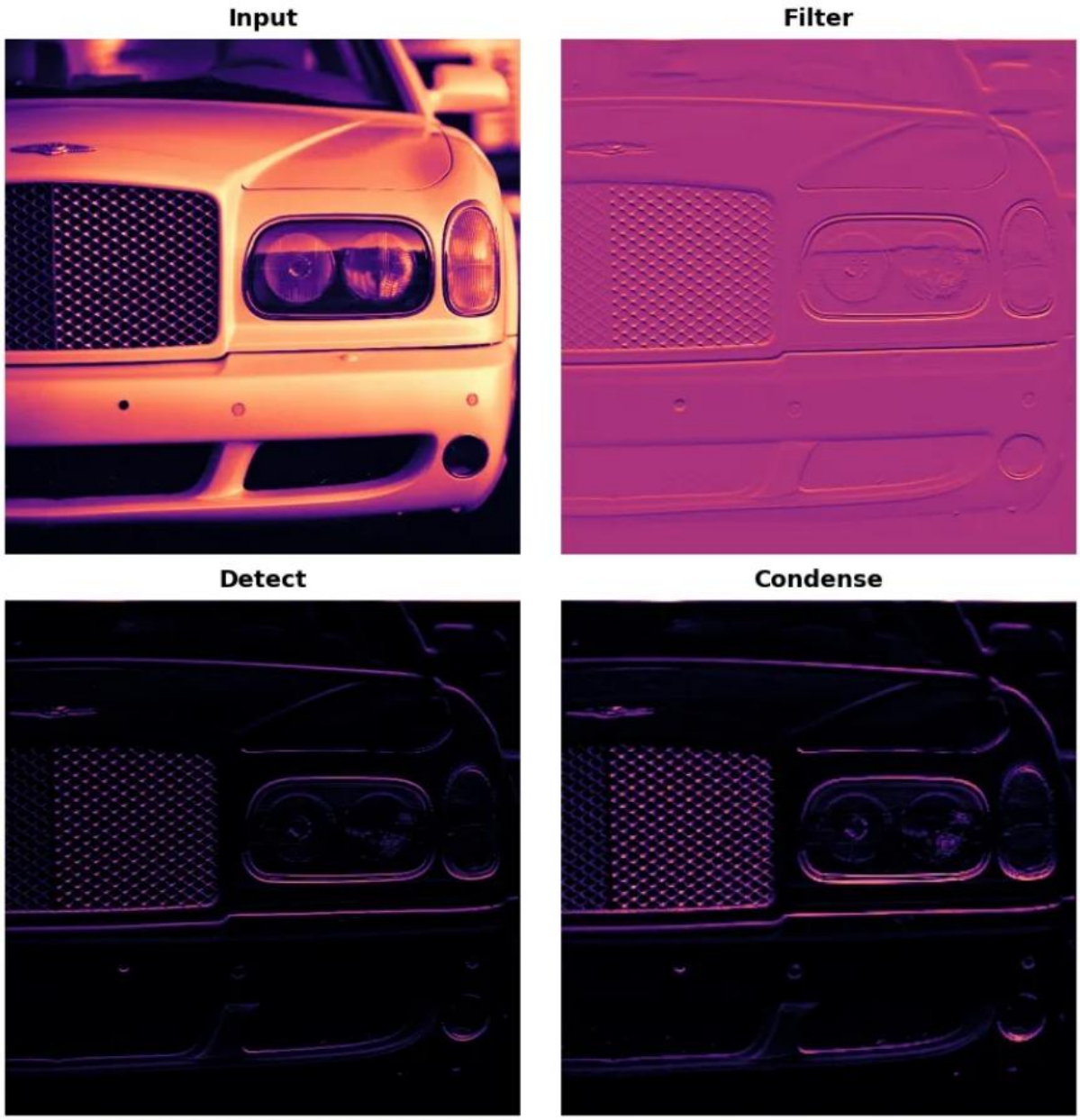


Figure 1: Visualization of convolution ("Detect") followed by max pooling ("Condense"). The pooling step compresses the strong activations while retaining their essential structure.

Source: Adapted from Danushi D. (2023), available at:
https://medium.com/@danushidk507/max-pooling-ef545993b6e4

Max Pooling Design Concept

Below is an example of a 4×4 input matrix being processed using a 2×2 max-pooling filter with a stride of 2.
Each 2×2 block is reduced to the single largest value in that block:

Input → Pooled Output → Final Max Value

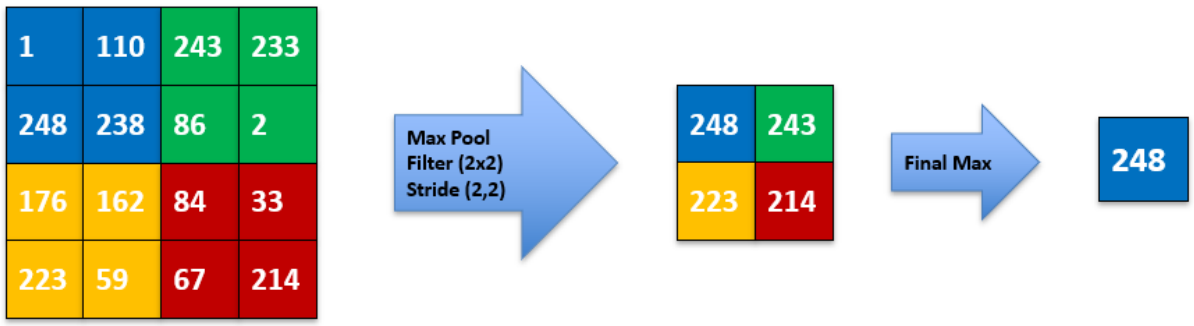


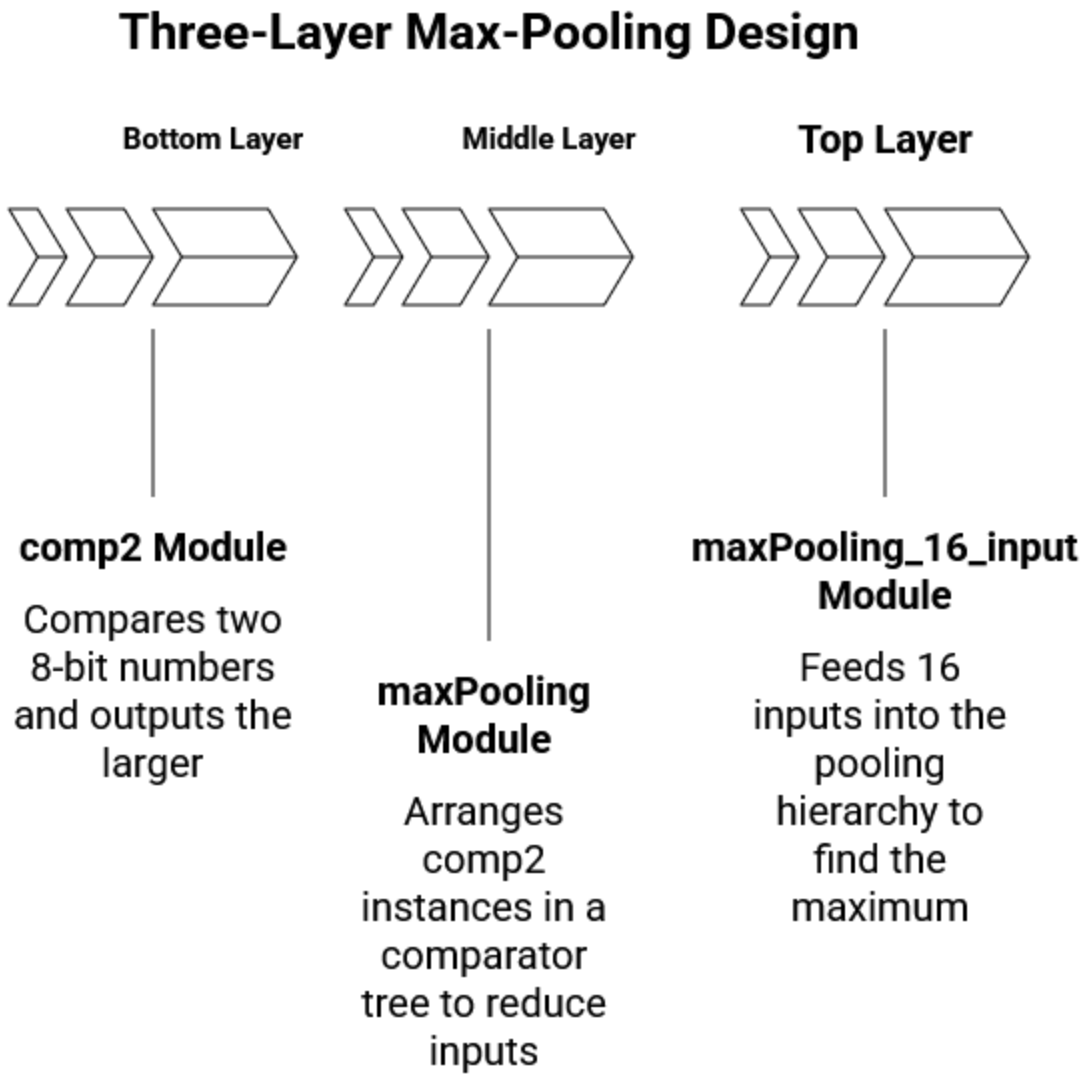
Figure 2: Max-pooling with a 2×2 filter and stride 2. Each coloured region shows the local window from which the maximum value is selected. The operation reduces the 4×4 matrix to a 2×2 matrix, and the global maximum among all regions is 248.

The entire design is built in three layers. At the bottom, the "comp2" module performs comparision task: it compares two 8-bit numbers and outputs the larger of the two. This module acts as the fundamental comparison block of the design.

In the next layer, the "maxPooling" module arranges several "comp2" instances in a reduction format, gradually eliminating inputs by comparing them in pairs. Essentially, it forms a comparator tree.

At the top level, "maxPooling_16_input" feeds sixteen inputs into the pooling hierarchy. The output of the top-level module is the maximum among all 16 inputs. This kind of max-pooling circuit is commonly used in hardware accelerators, image processing pipelines, and neural network architectures where multi-input reduction is required.

Max Pooling Design Concept



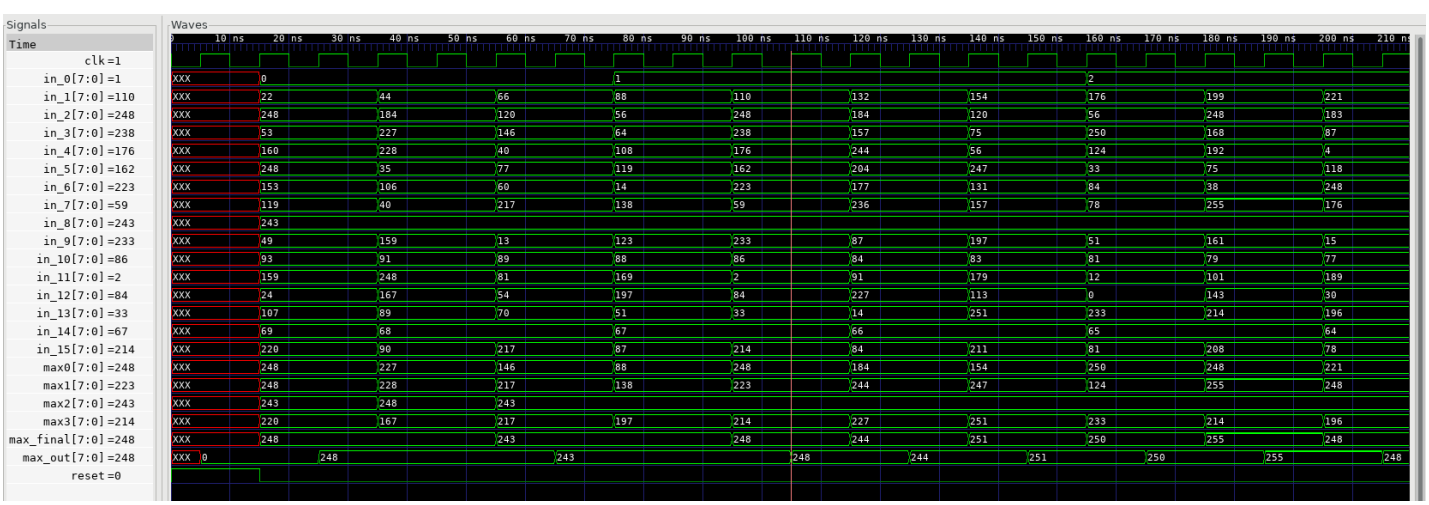
PART– 1 : PRE-SYNTHESIS

1. Screenshot of the output and gtkwave waveforms (pre-synthesis).

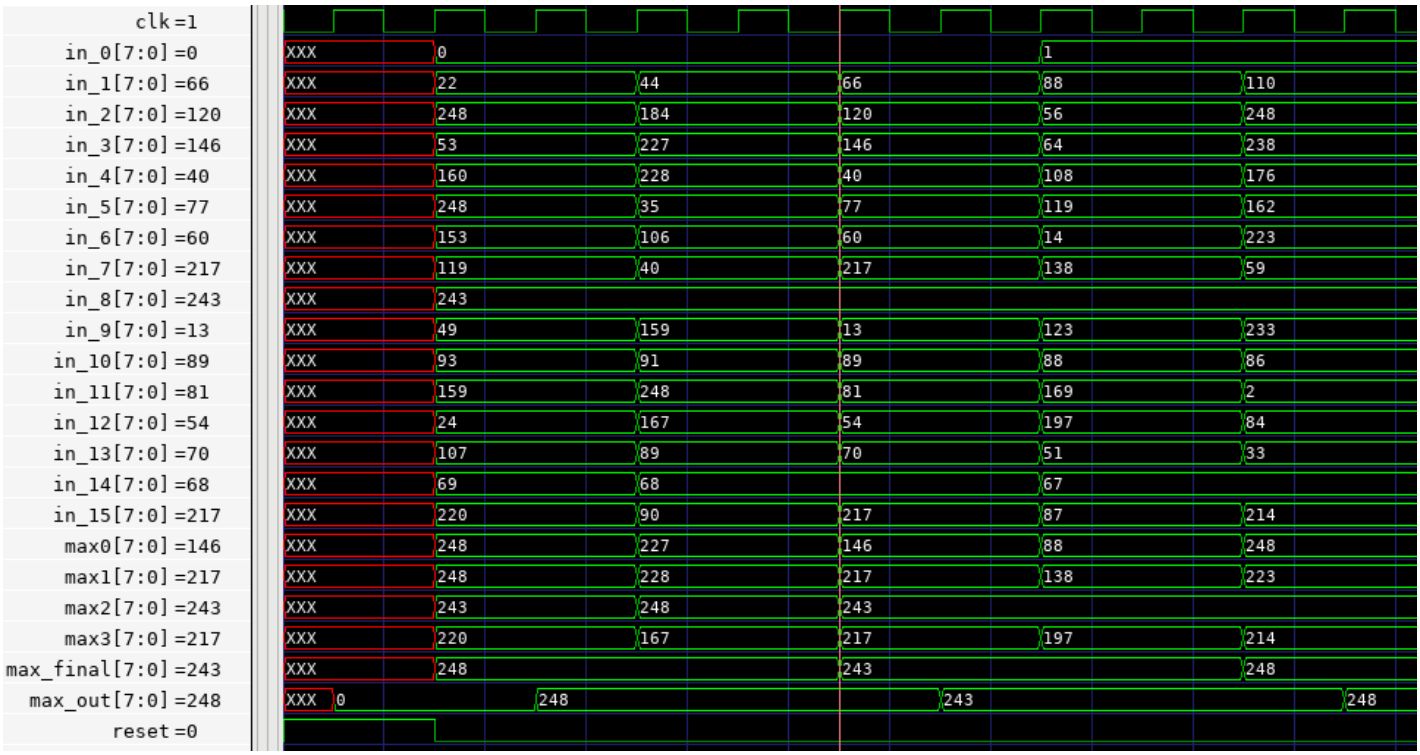
a) Output Transcript –

```
[EE671_45@vlsi73 Verilog]$ ./run-iverilog_top.sh
Testing: maxPooling_16_input
------------- MAXPOOL16 STRUCTURAL TEST WITH RESET -------------
VCD info: dumpfile waveforms_top.vcd opened for output.
PASS: Expected=248, DUT=248
PASS: Expected=248, DUT=248
PASS: Expected=243, DUT=243
PASS: Expected=243, DUT=243
PASS: Expected=248, DUT=248
PASS: Expected=244, DUT=244
PASS: Expected=251, DUT=251
PASS: Expected=250, DUT=250
PASS: Expected=255, DUT=255
PASS: Expected=248, DUT=248
------------- TEST FINISHED -------------
```

b) Output Waveform -



c) Output Waveform [Zoomed In] –



Note:

All the input and Output values are in Decimal.

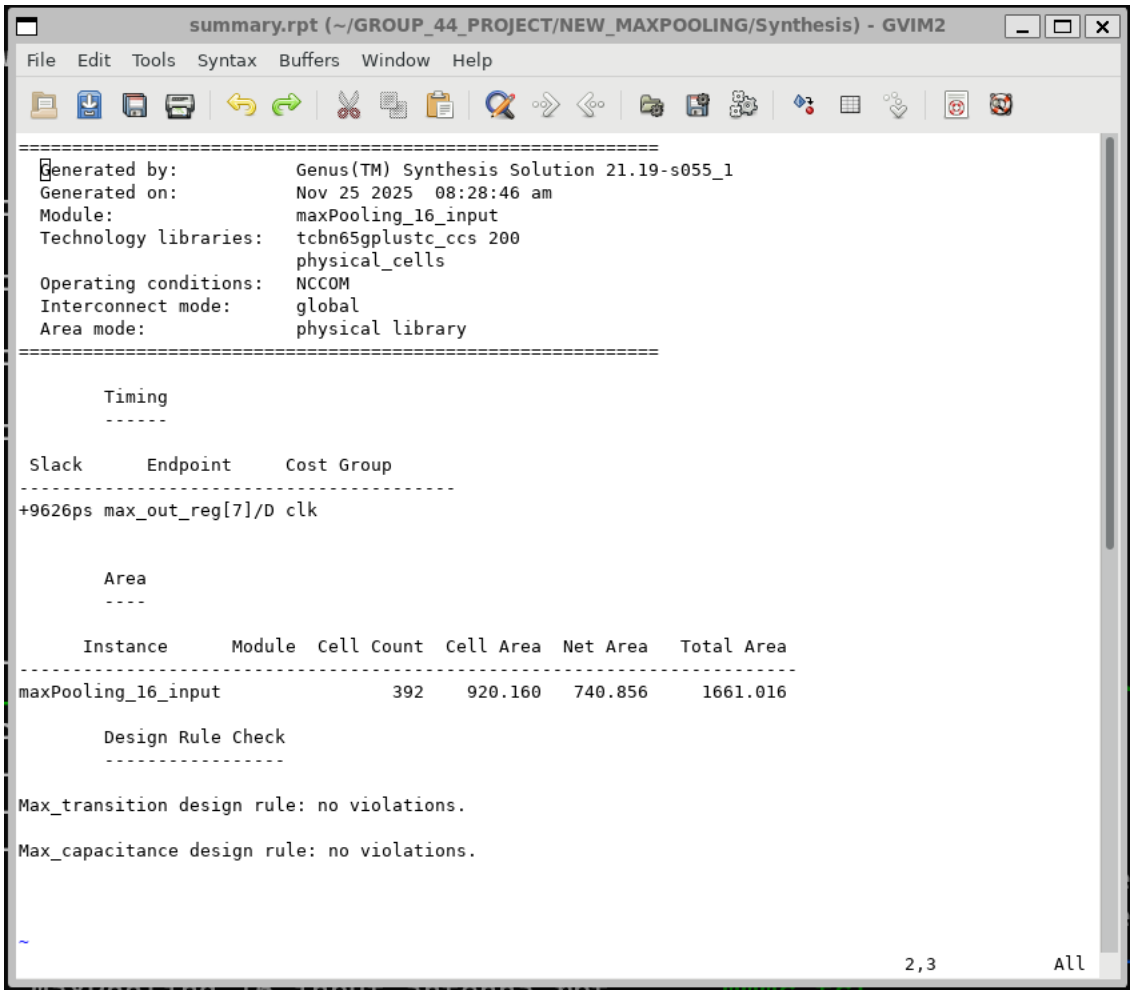
The Output (max_out signal) shows the maximum output across all 16 inputs.

This shows the outcome of the maxpool operation.

PART– 2 : SYNTHESIS

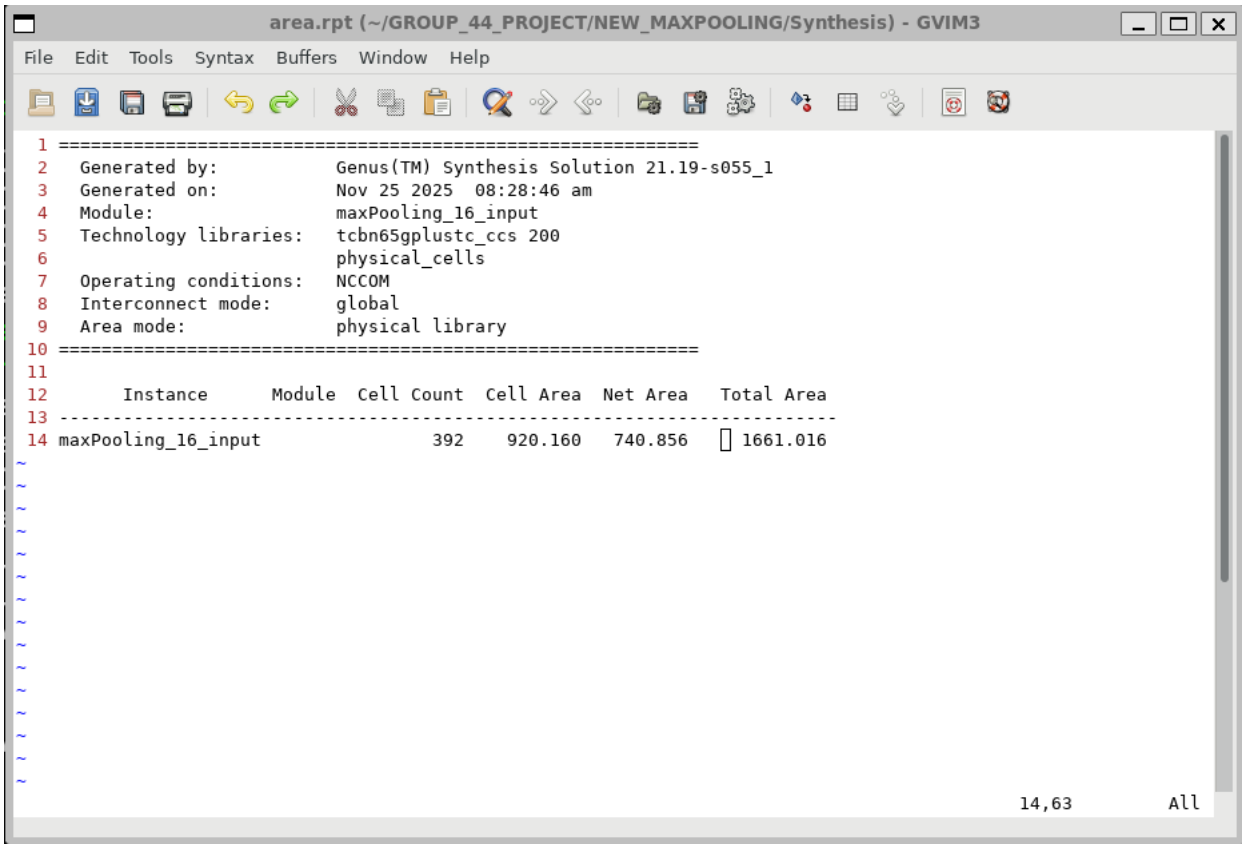
1. Screenshot of the area and summary report after synthesis.

a. Summary Report –



```
summary.rpt (~/GROUP_44_PROJECT/NEW_MAXPOOLING/Synthesis) - GVIM2
File  Edit  Tools  Syntax  Buffers  Window  Help

============================================================
  Generated by:           Genus(TM) Synthesis Solution 21.19-s055_1
  Generated on:           Nov 25 2025  08:28:46 am
  Module:                 maxPooling_16_input
  Technology libraries:   tcbn65gplustc_ccs 200
                          physical_cells
  Operating conditions:   NCCOM
  Interconnect mode:      global
  Area mode:              physical library
============================================================

        Timing
        ------

 Slack      Endpoint     Cost Group
----------------------------------------
+9626ps max_out_reg[7]/D clk


        Area
        ----

      Instance      Module  Cell Count  Cell Area  Net Area   Total Area
-----------------------------------------------------------------------------
maxPooling_16_input                392    920.160   740.856     1661.016

        Design Rule Check
        -----------------

Max_transition design rule: no violations.

Max_capacitance design rule: no violations.


~
                                                                  2,3          All
```

b. Area Report –




```
============================================================
  Generated by:           Genus(TM) Synthesis Solution 21.19-s055_1
  Generated on:           Nov 25 2025  08:28:46 am
  Module:                 maxPooling_16_input
  Technology libraries:   tcbn65gplustc_ccs 200
                          physical_cells
  Operating conditions:   NCCOM
  Interconnect mode:      global
  Area mode:              physical library
============================================================

      Instance      Module  Cell Count  Cell Area  Net Area   Total Area
-----------------------------------------------------------------------
maxPooling_16_input                392    920.160   740.856   1661.016
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
                                                                14,63         All
```

c. Gates Report –

gates.rpt (~/GROUP_44_PROJECT/NEW_MAXPOOLING/Synthesis) - GVIM4

File Edit Tools Syntax Buffers Window Help

```
============================================================
  Generated by:           Genus(TM) Synthesis Solution 21.19-s055_1
  Generated on:           Nov 25 2025  08:28:46 am
  Module:                 maxPooling_16_input
  Technology libraries:   tcbn65gplustc_ccs 200
                          physical_cells
  Operating conditions:   NCCOM
  Interconnect mode:      global
  Area mode:              physical library
============================================================


   Gate    Instances   Area          Library
--------------------------------------------------
AO211D0            2    5.760    tcbn65gplustc_ccs
AO21D0             1    2.520    tcbn65gplustc_ccs
AO22D0             1    2.880    tcbn65gplustc_ccs
AOI211D0           1    2.520    tcbn65gplustc_ccs
AOI21D0            3    6.480    tcbn65gplustc_ccs
AOI221D0           1    2.880    tcbn65gplustc_ccs
AOI222D0           3   10.800    tcbn65gplustc_ccs
AOI22D0            6   15.120    tcbn65gplustc_ccs
AOI32D0           11   31.680    tcbn65gplustc_ccs
CKAN2D1           10   21.600    tcbn65gplustc_ccs
CKND0            104  112.320    tcbn65gplustc_ccs
CKND2D0           11   15.840    tcbn65gplustc_ccs
DFQD1              8   54.720    tcbn65gplustc_ccs
FCICIND1          10   46.800    tcbn65gplustc_ccs
IND2D0             3    6.480    tcbn65gplustc_ccs
INR2D0             3    6.480    tcbn65gplustc_ccs
INR2XD0            1    2.160    tcbn65gplustc_ccs
INVD1              2    2.160    tcbn65gplustc_ccs
MAOI222D0         33   95.040    tcbn65gplustc_ccs
MAOI22D0           2    5.760    tcbn65gplustc_ccs
MOAI22D0           8   23.040    tcbn65gplustc_ccs
MUX2D0             1    3.240    tcbn65gplustc_ccs
MUX2ND0           89  256.320    tcbn65gplustc_ccs
MUX3ND0            4   21.600    tcbn65gplustc_ccs
NR2D0             20   28.800    tcbn65gplustc_ccs
NR2XD0             9   12.960    tcbn65gplustc_ccs
OA32D0             1    3.600    tcbn65gplustc_ccs
OAI21D1            3    6.480    tcbn65gplustc_ccs
OAI221D1           1    3.240    tcbn65gplustc_ccs
OAI222D0           9   29.160    tcbn65gplustc_ccs
OAI222D1           2    7.920    tcbn65gplustc_ccs
OAI22D0            2    5.040    tcbn65gplustc_ccs
OAI22D1            7   17.640    tcbn65gplustc_ccs
OAI32D0           11   31.680    tcbn65gplustc_ccs
OR2D0              9   19.440    tcbn65gplustc_ccs
--------------------------------------------------
total            392  920.160
```

52,0-1 Top

```
52
53
54
55      Type       Instances   Area  Area %
56 ----------------------------------------
57 sequential             8  54.720    5.9
58 inverter             106 114.480   12.4
59 logic                278 750.960   81.6
60 physical_cells         0   0.000    0.0
61 ----------------------------------------
62 total                392 920.160  100.0
63
~
                                                                49,51         Bot
```

d. Power Report –

```
power.rpt (~/GROUP_44_PROJECT/NEW_MAXPOOLING/Synthesis) - GVIM5
File  Edit  Tools  Syntax  Buffers  Window  Help

Instance: /maxPooling_16_input
Power Unit: W
PDB Frames: /stim#0/frame#0
  ----------------------------------------------------------------------
    Category         Leakage     Internal    Switching        Total    Row%
  ----------------------------------------------------------------------
       memory     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
     register     2.82183e-07  1.25810e-05  6.80000e-07  1.35432e-05  12.36%
        latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
        logic     3.05242e-06  3.97645e-05  5.26408e-05  9.54577e-05  87.13%
         bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
        clock     0.00000e+00  0.00000e+00  5.60000e-07  5.60000e-07   0.51%
          pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
           pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
  ----------------------------------------------------------------------
    Subtotal     3.33460e-06  5.23456e-05  5.38808e-05  1.09561e-04 100.00%
  Percentage           3.04%       47.78%       49.18%      100.00% 100.00%
  ----------------------------------------------------------------------
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
                                                                3,1           All
```

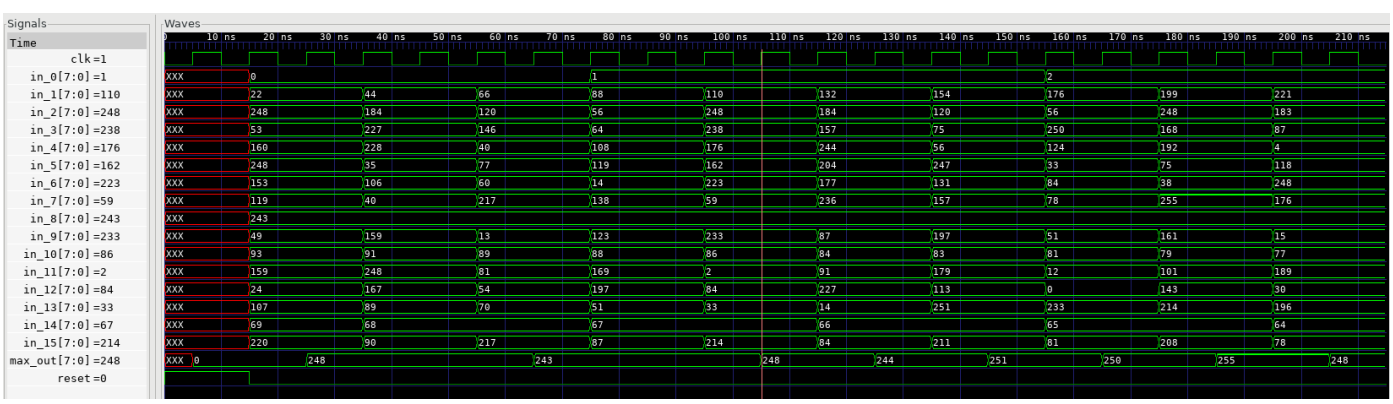
---

2. Screenshot of the output and gtkwave waveform (post-synthesis).

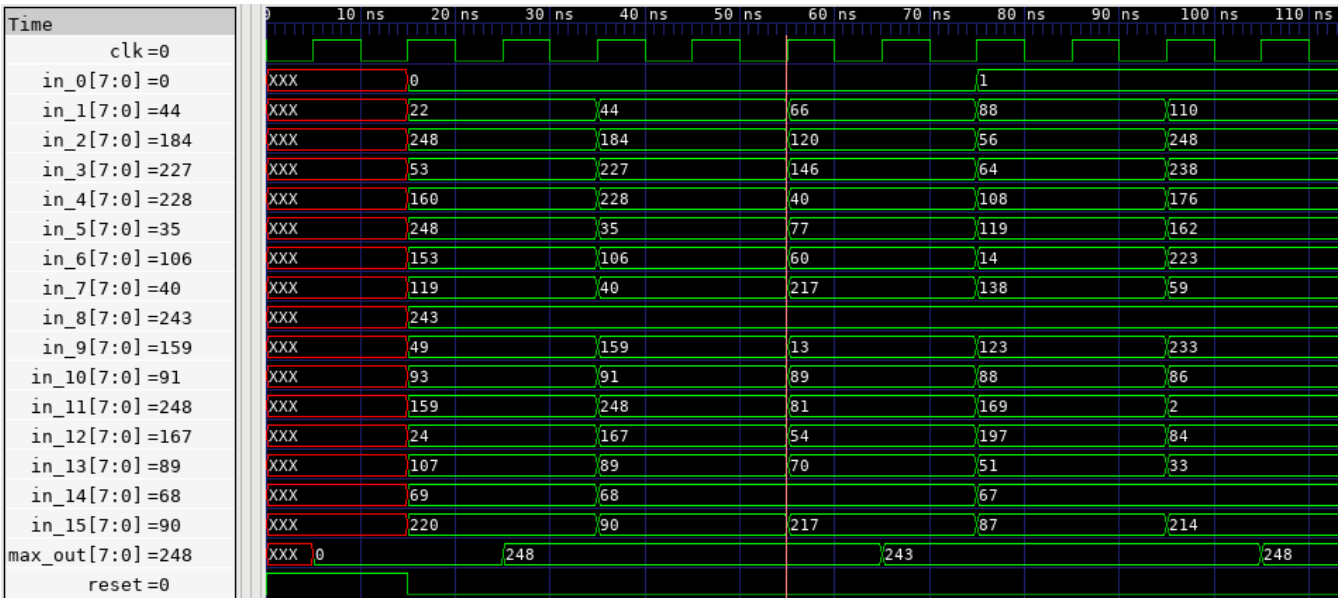
a) Output Transcript –

```
ncsim> source /vlsi/cad/cadence/INCISIVE152/tools/inca/files/ncsimrc
ncsim> run
------------- MAXPOOL16 STRUCTURAL TEST WITH RESET -------------
PASS: Expected=248, DUT=248
PASS: Expected=248, DUT=248
PASS: Expected=243, DUT=243
PASS: Expected=243, DUT=243
PASS: Expected=248, DUT=248
PASS: Expected=244, DUT=244
PASS: Expected=251, DUT=251
PASS: Expected=250, DUT=250
PASS: Expected=255, DUT=255
PASS: Expected=248, DUT=248
------------- TEST FINISHED -------------
Simulation complete via $finish(1) at time 215 NS + 0
./maxPooling_16_input_TB.v:94         $finish;
ncsim> exit
[EE671_45@vlsi73 Verification]$
```

b) Output Waveform -



c) Output Waveform [Zoomed In] –



Note:

All the input and Output values are in Decimal.

The Output (max_out signal) shows the maximum output across all 16 inputs.

This shows the outcome of the maxpool operation.

---

3. The following table generated from the RTL file:

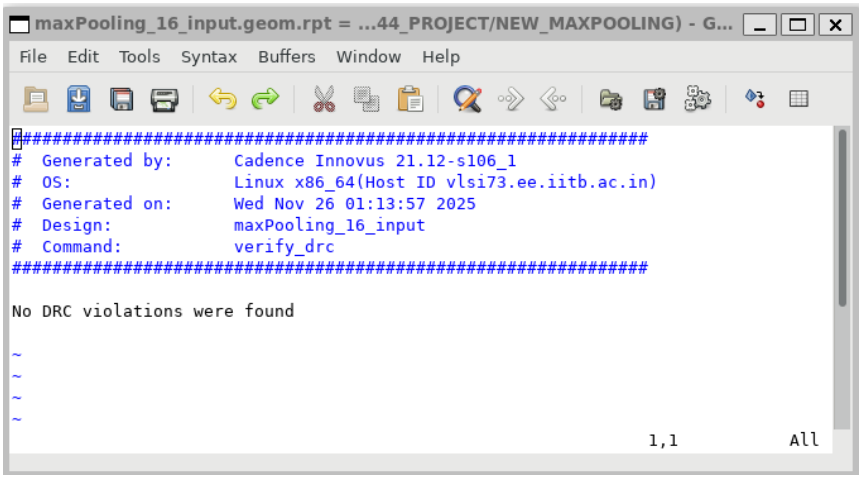
Table –

Number of combinational cells	384 (Cell Area: 865.44)
Number of Sequential cells (Flip-flops)	08 (Cell Area: 54.72)
Total number of cells	392 (Cell Area: 920.16) (Net Area: 740.856) (Total Area: 1661.016)

PART – 3 : PHYSICAL DESIGN

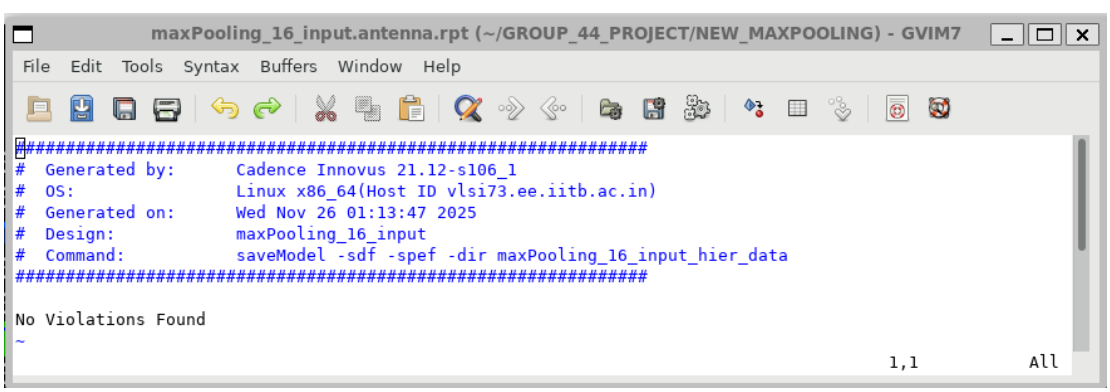
1. Screenshot of the DRC, Antenna, Connectivity (for both signal and special route) reports showing 0 violations.

a) DRC Check –



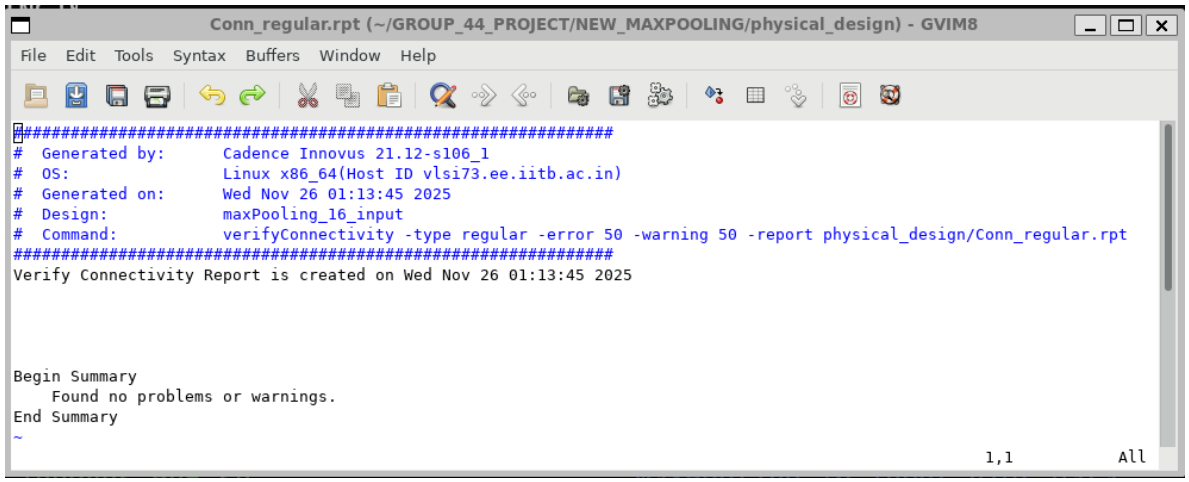
DRC Check

b) Antenna Violation Check –



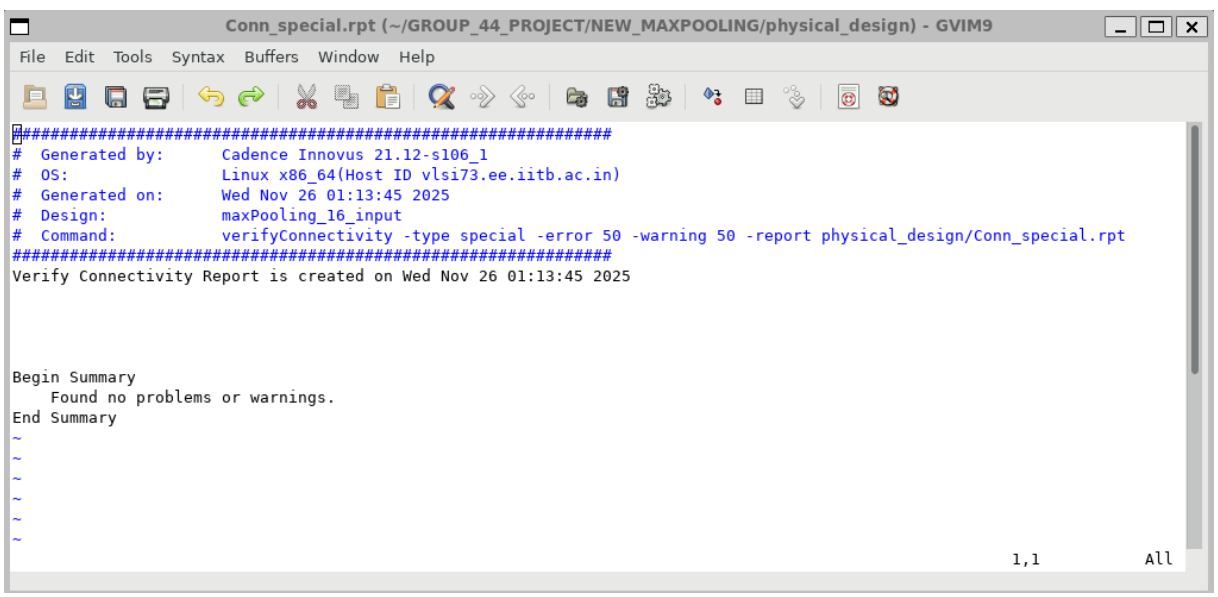
Antenna Violation Check

c) Connectivity Report (Regular) –



Regular Connectivity Report (for Signal)

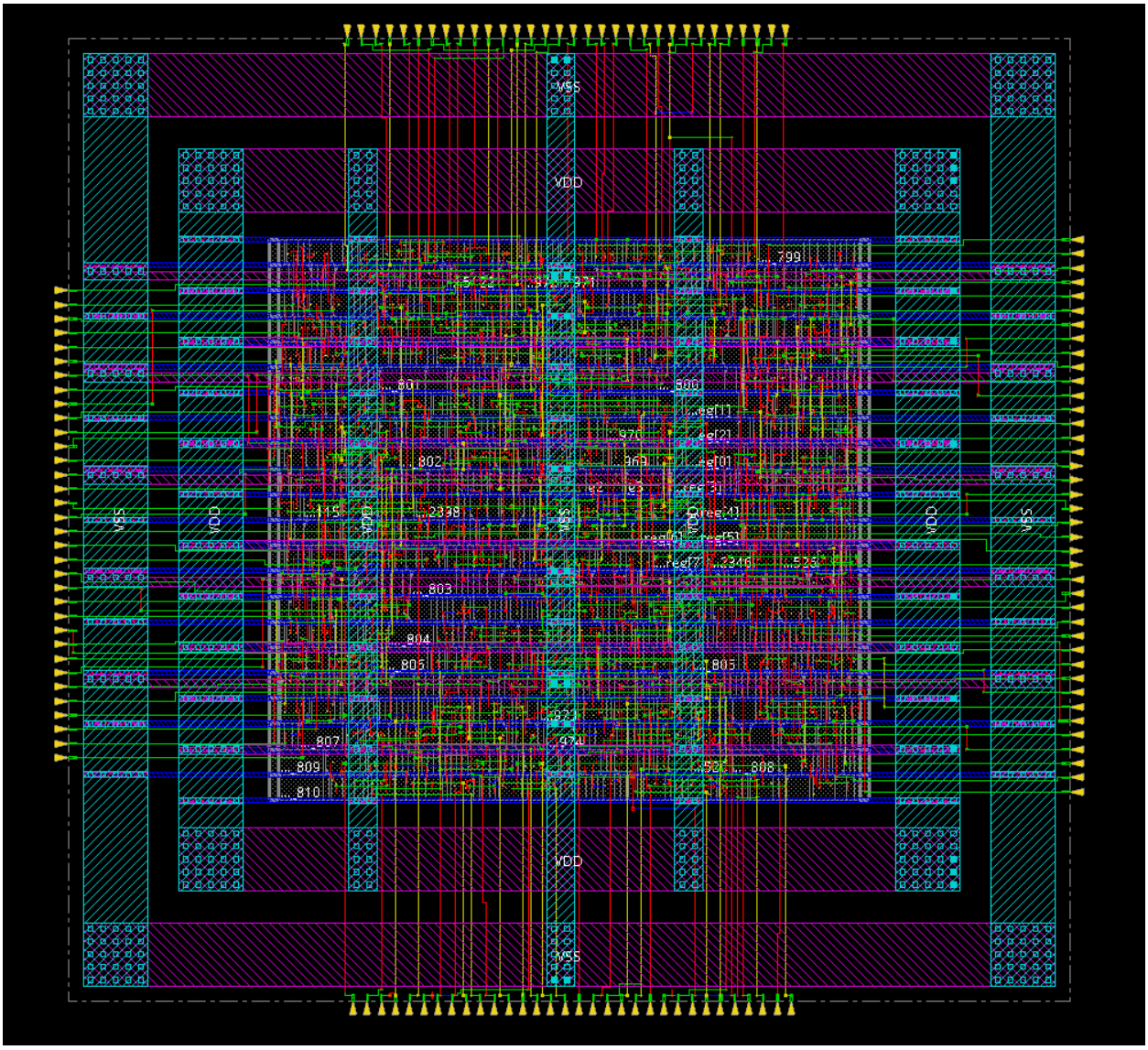
d) Connectivity Report (Special) –



Connectivity Report (for Special Route)

2. Screenshot of the final Layout from the Innovus window.

The design is PNR-heavy. The design has a total of 138 pins, which are distributed over the chip/die area.



Final Layout

---

3. **Following information from the post route timing, area and power reports:**

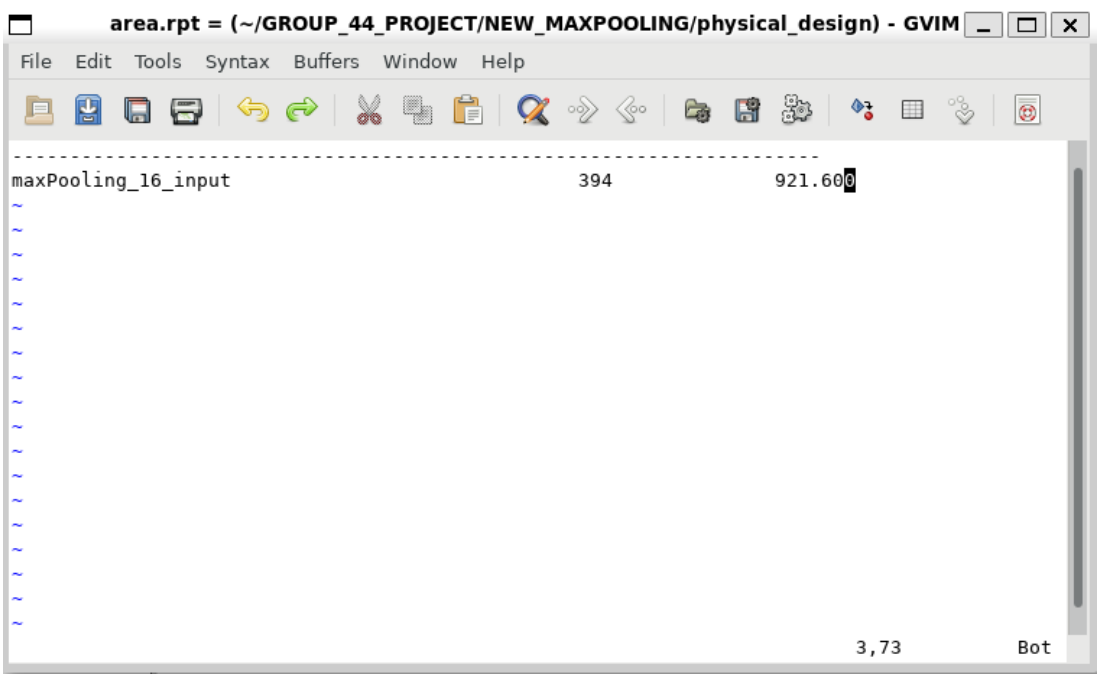
The following values are listed from the Post Route summary files dumped in the repo.

Table –

Clock Frequency (MHz)	**100 MHz**
Worst case setup slack [WNS in the report] (ns)	**6.321 ns**
Worst case hold slack [WNS in the report] (ns)	**0.039 ns**
Design area (µm2)	**921.600 um2 (Post PD)**
Power Consumption (Sequential Only) (µW)	**16.4 uW**
Power Consumption (Combinational Only) (µW)	**109.2 uW**
Total Power Consumption (Internal Only) (µW)	**68.85 uW**

Total Power Consumption (Switching Only) (µW)	40.26 uW
Total Power Consumption (Leakage Only) (µW)	27.24 uW
Total Power Consumption (µW)	136.3 uW

a) Area Report –



PNR Area Report

b) Setup Time Report –

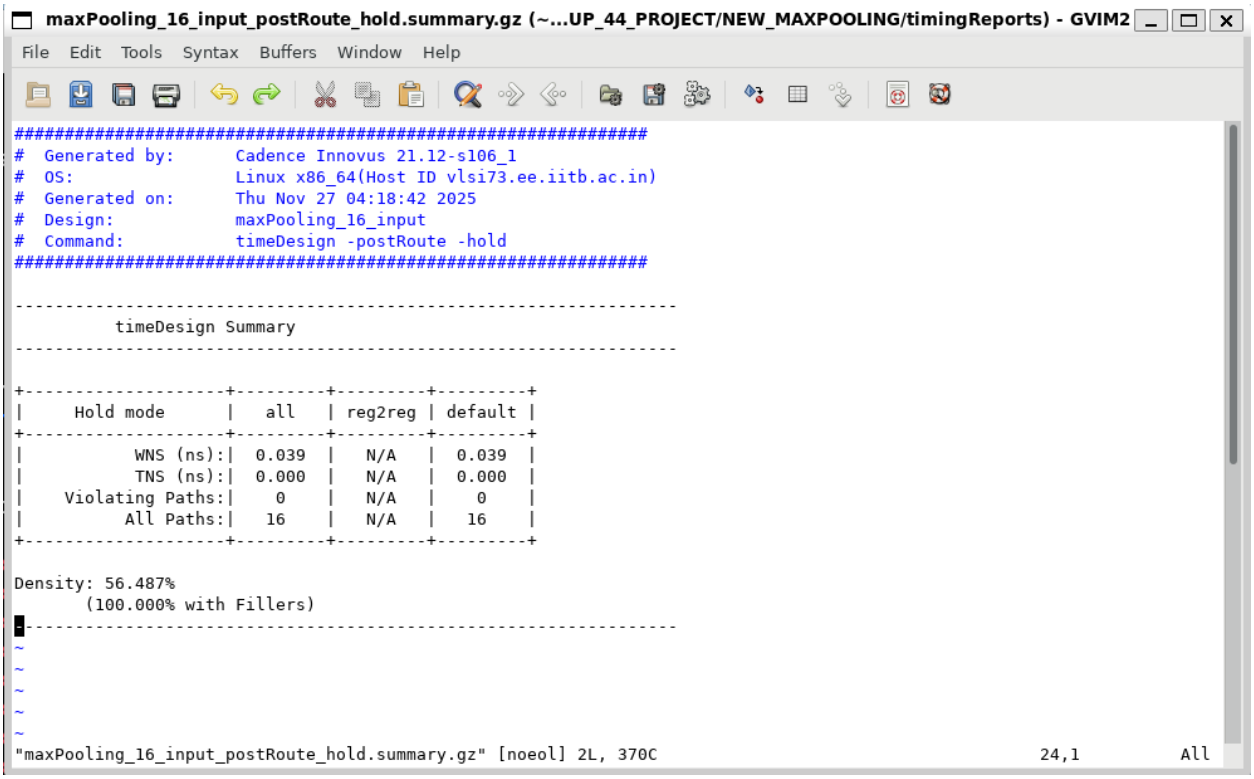
maxPooling_16_input_postRoute.summary.gz (~/GROUP_44_PROJECT/NEW_MAXPOOLING/timingReports) - GVIM1

File Edit Tools Syntax Buffers Window Help

```
#############################################################
#  Generated by:      Cadence Innovus 21.12-s106_1
#  OS:                Linux x86_64(Host ID vlsi73.ee.iitb.ac.in)
#  Generated on:      Thu Nov 27 04:18:40 2025
#  Design:            maxPooling_16_input
#  Command:           timeDesign -postRoute
#############################################################

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  6.321  |   N/A   |  6.321  |
|           TNS (ns):|  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |   N/A   |    0    |
|          All Paths:|   16    |   N/A   |   16    |
+--------------------+---------+---------+---------+

+----------------+-----------------------------+------------------+
|                |              Real           |       Total      |
|    DRVs        +------------------+----------+------------------|
|                |  Nr nets(terms)  | Worst Vio |  Nr nets(terms)  |
+----------------+------------------+----------+------------------+
|   max_cap      |      0 (0)       |   0.000  |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000  |      0 (0)       |
|   max_fanout   |      0 (0)       |     0    |      0 (0)       |
|   max_length   |      0 (0)       |     0    |      0 (0)       |
+----------------+------------------+----------+------------------+

Density: 56.487%
       (100.000% with Fillers)
------------------------------------------------------------------
~
                                                                    4,1           All
```

Post Route Timing Summary

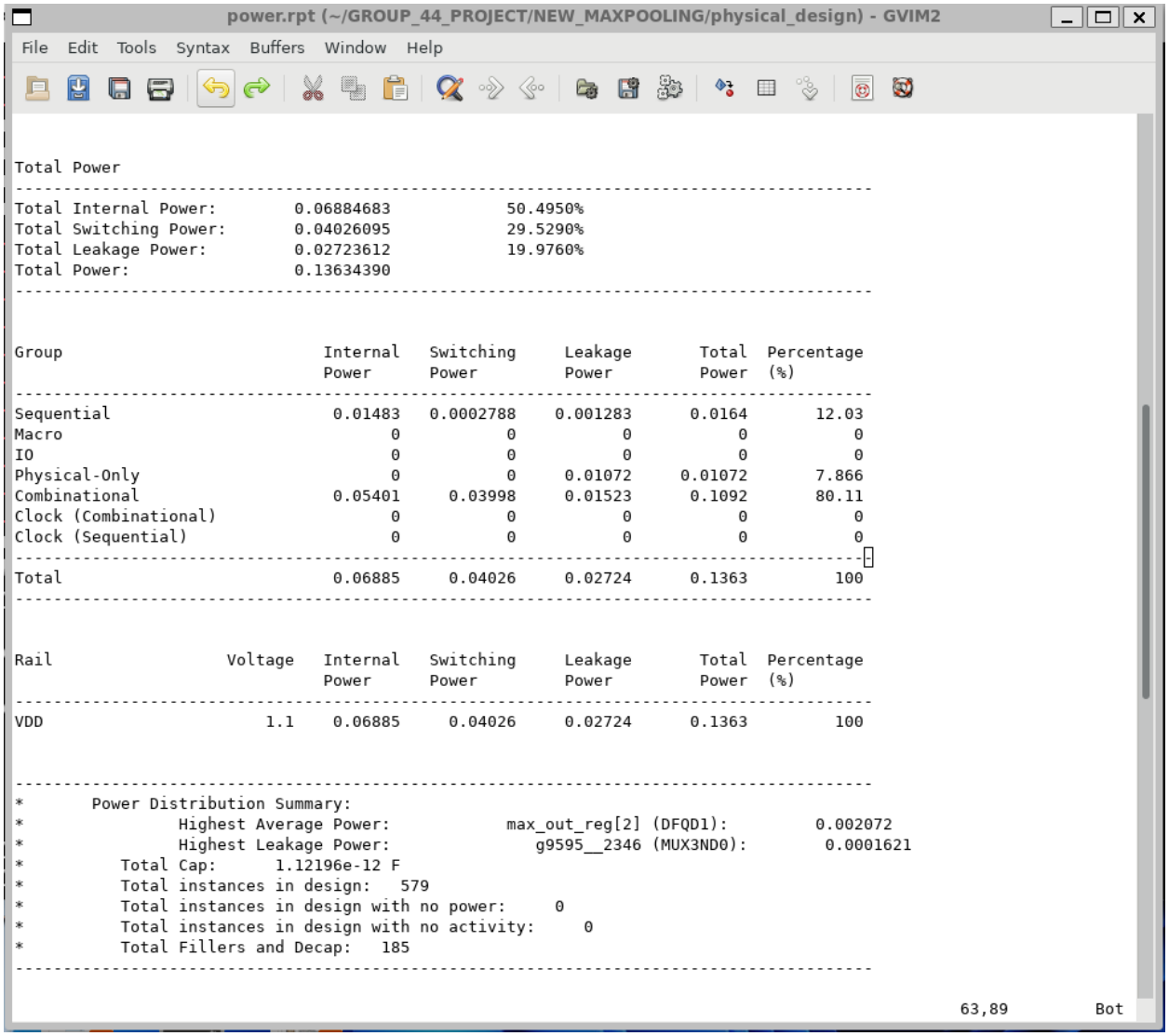
c) Hold Time Report –



```
maxPooling_16_input_postRoute_hold.summary.gz (~...UP_44_PROJECT/NEW_MAXPOOLING/timingReports) - GVIM2
File  Edit  Tools  Syntax  Buffers  Window  Help

#############################################################
#  Generated by:      Cadence Innovus 21.12-s106_1
#  OS:                Linux x86_64(Host ID vlsi73.ee.iitb.ac.in)
#  Generated on:      Thu Nov 27 04:18:42 2025
#  Design:            maxPooling_16_input
#  Command:           timeDesign -postRoute -hold
#############################################################

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.039  |   N/A   |  0.039  |
|           TNS (ns):|  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |   N/A   |    0    |
|          All Paths:|   16    |   N/A   |   16    |
+--------------------+---------+---------+---------+

Density: 56.487%
       (100.000% with Fillers)
------------------------------------------------------------------
~
~
~
~
~
"maxPooling_16_input_postRoute_hold.summary.gz" [noeol] 2L, 370C                    24,1          All
```

Post Route Hold Summary

d) Power Report –




```
Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:        0.06884683            50.4950%
Total Switching Power:       0.04026095            29.5290%
Total Leakage Power:         0.02723612            19.9760%
Total Power:                 0.13634390
-----------------------------------------------------------------------------------------


Group                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
Sequential                       0.01483   0.0002788    0.001283      0.0164       12.03
Macro                                  0           0           0           0           0
IO                                     0           0           0           0           0
Physical-Only                          0           0     0.01072     0.01072       7.866
Combinational                    0.05401     0.03998     0.01523      0.1092       80.11
Clock (Combinational)                  0           0           0           0           0
Clock (Sequential)                     0           0           0           0           0
-----------------------------------------------------------------------------------------
Total                            0.06885     0.04026     0.02724      0.1363         100
-----------------------------------------------------------------------------------------


Rail                  Voltage   Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
VDD                       1.1    0.06885     0.04026     0.02724      0.1363         100


-----------------------------------------------------------------------------------------
*       Power Distribution Summary:
*               Highest Average Power:            max_out_reg[2] (DFQD1):         0.002072
*               Highest Leakage Power:               g9595__2346 (MUX3ND0):        0.0001621
*          Total Cap:      1.12196e-12 F
*          Total instances in design:   579
*          Total instances in design with no power:     0
*          Total instances in design with no activity:     0
*          Total Fillers and Decap:   185
-----------------------------------------------------------------------------------------
                                                                                 63,89         Bot
```

Power Report

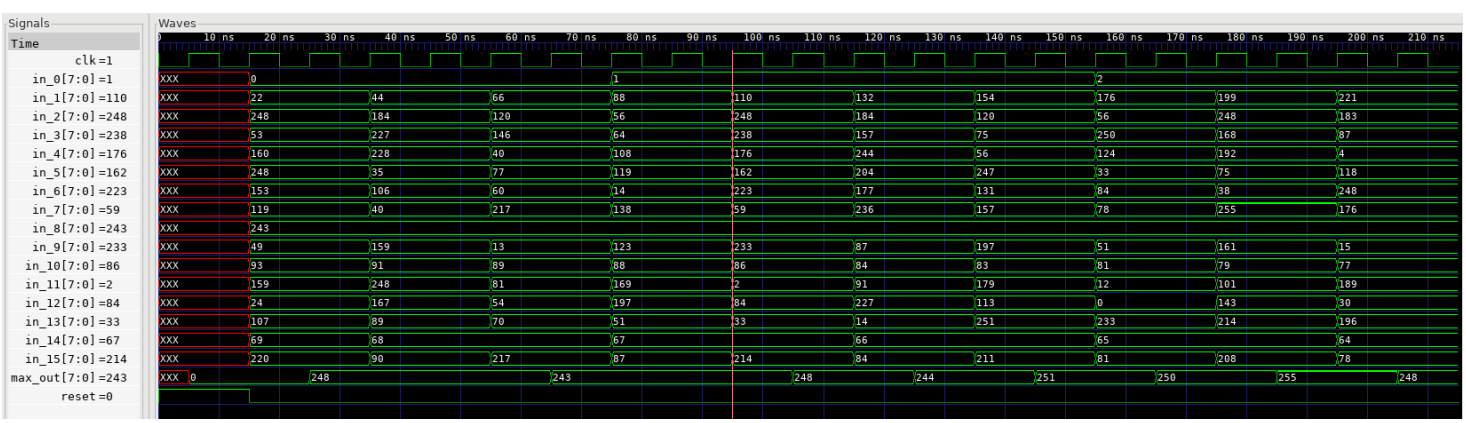
---

4. **Verify the post physical design RTL with a functional simulation using the same testbench that was used to verify the post synthesis RTL in assignment-3. Provide a screenshot of the Gtkwave waveform dump.**

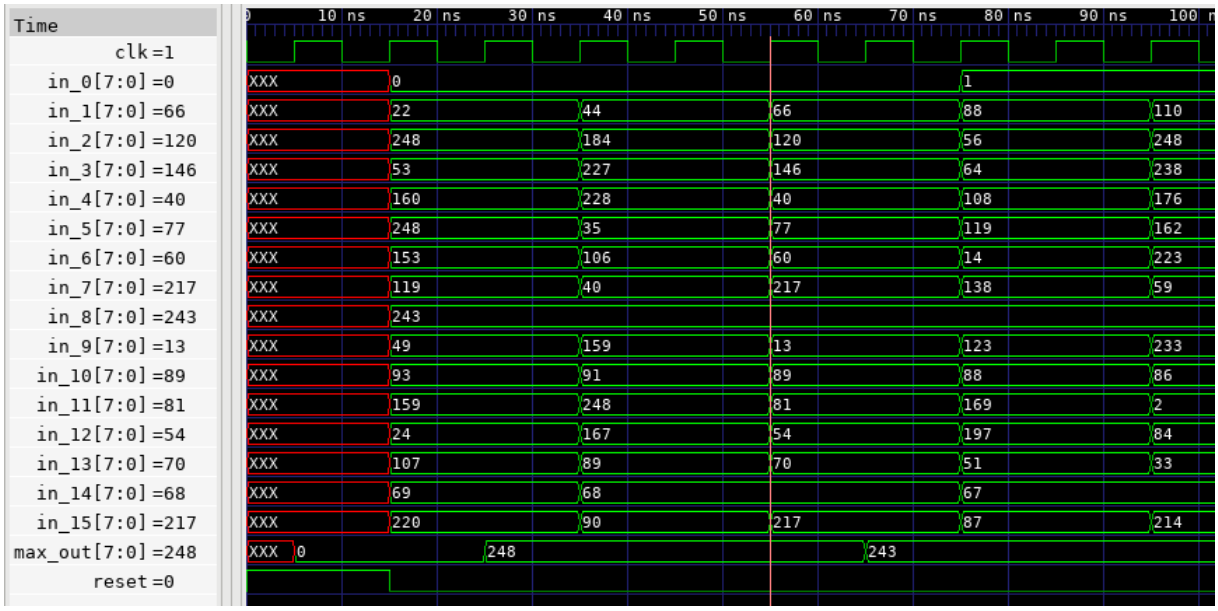
a) Output Transcript (Post PNR) –

```
ncsim> run
------------- MAXPOOL16 STRUCTURAL TEST WITH RESET -------------
PASS: Expected=248, DUT=248
PASS: Expected=248, DUT=248
PASS: Expected=243, DUT=243
PASS: Expected=243, DUT=243
PASS: Expected=248, DUT=248
PASS: Expected=244, DUT=244
PASS: Expected=251, DUT=251
PASS: Expected=250, DUT=250
PASS: Expected=255, DUT=255
PASS: Expected=248, DUT=248
------------- TEST FINISHED -------------
Simulation complete via $finish(1) at time 215 NS + 0
./maxPooling_16_input_TB.v:94         $finish;
ncsim> exit
```

b) Output Waveform (Post PNR) –



c) Output Waveform [Zoomed In] –



Note:

All the input and Output values are in Decimal.

The Output (max_out signal) shows the maximum output across all 16 inputs.

This shows the outcome of the maxpool operation.

Table of Contributions

Contribution Summary Information –

Topic	Contributors
Design Selection	Karunesh Kumar Tripathi and Tejas Hande
Verilog Implementation	Karunesh Kumar Tripathi and Tejas Hande
Verification by Simulation	Karunesh Kumar Tripathi and Tejas Hande
Synthesis	Amit Kale and Antriksh Thakur
Post Synthesis Simulation	Amit Kale and Antriksh Thakur
Physical Design	Antriksh Thakur
Physical Design Debug	Tejas Hande
Physical Design Verification	Antriksh Thakur, Karunesh Kumar Tripathi and Tejas Hande
Post PD Simulation	Amit Kale and Antriksh Thakur
Report Preperation	Amit Kale, Antriksh Thakur, Karunesh Kumar Tripathi and Tejas Hande